



Feeding Attendance

The Impact of California's Universal Meals Program on
Public School Absenteeism

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Introduction

Motivation

- 8 states adopted universal free meal policies for K-12 students in the past 2 years.
- **Advocates:** Equal access to food, reduced stigma, improved academics, and reduced food insecurity.
- **Critics:** Increased taxpayer costs, especially in wealthier districts with lower food insecurity.

This begs the question:

Can universal free meals deliver meaningful benefits across socioeconomic contexts?

Motivation (cont.)



- Universal Meals Program (UMP)
 - First implemented in California during 2022-2023 academic year
 - Free meals to all K-12 public school students, regardless of income
- Free meals may reduce hunger-related absences (Frisvold, 2015).
- Chronic absenteeism
 - Defined as missing 10% or more of the school year
 - Linked to lower academic performance, higher dropout rates, and increased juvenile delinquency risks

Free Meals

improve attendance,
class participation,
and student health.

(Schwartz & Rothbart, 2019)

Chronic absenteeism

is an early predictor
of dropout.

(Balfanz & Byrnes, 2012)

Criminal activity

is more prevalent
among dropouts.

(Lochner & Moretti, 2004;
Deming, 2011)

A bold assumption...

no meals → chronically absent → **prison**

Data

- California Department of Education (CDE)
 - Chronic absenteeism rates for K-12 public schools
 - Free and Reduced-Price Meals (FRPM) eligibility
- 7 academic years from 2016-17 to 2023-24
 - 2019-20 excluded due to absence of data from COVID-19 pandemic
 - 2022-23 and 2023-24 treatment periods
- Exclude districts w/o complete data of all 7 academic years and continuation and special education schools
- District-level panel data
 - 972 districts
 - 6,804 observations

Data Descriptions

- District poverty levels

Low $\leq 25\%$

Mid-Low $> 25\% - \leq 50\%$

Mid-High $> 50\% - \leq 75\%$

High $> 75\%$

(defined by the National Center for Education Statistics)

- Demographics

Gender* female, male

Race Asian, Black, Hispanic, White

Subgroups disabled, foster, homeless

*not all genders listed due to incomplete information across academic years

Example

This is my former school district (!!!):

academic...	districtcode	districtname	GF	GM	RA	RB	RH	RW	SD	SF	SH	TA	fmk12	fpmk12	poverty...
2016-17	75291	San Gabriel Unified	.0870671	.0997072	.0262443	.1976048	.121755	.1196911	.1820809	.2884615	.2146597	.0934297	.435576	.5706125	3
2017-18	75291	San Gabriel Unified	.1035266	.1109327	.033574	.2145923	.1203008	.1520894	.2273524	.28	.3434343	.1071689	.5452788	.6398399	2
2018-19	75291	San Gabriel Unified	.0725016	.0803717	.0269856	.1620112	.1180672	.1032807	.1587537	.2045455	.3154362	.0764399	.4842402	.5840817	3
2020-21	75291	San Gabriel Unified	.0988235	.0905647	.0127186	.1707317	.1658643	.2040134	.1612378	.2954545	.4259259	.095092	.4966455	.571748	3
2021-22	75291	San Gabriel Unified	.2075271	.1979011	.0802693	.40825	.3295533	.299389	.3721028	.3595506	.6208531	.2027862	.5010889	.6458127	2
2022-23	75291	San Gabriel Unified	.219906	.2045583	.0727893	.5	.3415941	.2963351	.3370166	.4761905	.4516906	.2126159	.5411264	.6803079	2
2023-24	75291	San Gabriel Unified	.1740593	.1529568	.0621054	.2682927	.2790251	.2139973	.2969617	.4133333	.3662651	.1638726	.5442149	.6684296	2

For analysis, each district's poverty level is fixed based on its classification in the 2021-22 academic year (the last year of the pre-treatment period).

Summary Statistics

Table 1: Chronic Absenteeism Rate by Demographic Across Academic Years

	2016-17	2017-18	2018-19	2020-21	2021-22	2022-23	2023-24	Average (%)
Female	.125	.127	.126	.145	.289	.245	.199	17.98%
Male	.127	.127	.129	.163	.293	.252	.204	18.54%
Asian	.059	.064	.063	.070	.162	.137	.110	9.54%
Black	.171	.175	.181	.220	.353	.304	.266	23.83%
Hispanic	.130	.132	.132	.171	.315	.269	.220	19.59%
White	.128	.129	.126	.137	.273	.238	.190	17.47%
Disabled	.174	.177	.178	.207	.369	.319	.265	24.21%
Foster	.256	.257	.253	.330	.435	.374	.352	31.88%
Homeless	.277	.284	.283	.347	.481	.424	.371	35.42%
Total	.127	.128	.127	.156	.293	.253	.203	18.43%

Summary Statistics (cont.)

Table 2: District Counts by Poverty Level Across Academic Years

	2016-17	2017-18	2018-19	2020-21	2021-22	2022-23	2023-24	Total
Low Poverty	198	147	151	163	153	136	120	1068 (15.70%)
Mid-Low Poverty	344	322	326	325	346	335	321	2,319 (34.08%)
Mid-High Poverty	180	246	237	229	234	249	238	1,613 (23.71%)
High Poverty	250	257	258	255	239	252	293	1,804 (25.93%)
Total	972	972	972	972	972	972	972	6,804 (100.00%)

Empirical Model

$$\begin{aligned}\text{Absenteeism}_{it} = & \beta_0 + \beta_1(\text{Post}_t) + \beta_2(\text{Post}_t \cdot \text{MidLowPoverty}_i) \\ & + \beta_3(\text{Post}_t \cdot \text{MidHighPoverty}_i) \\ & + \beta_4(\text{Post}_t \cdot \text{HighPoverty}_i) + \delta_i + \gamma_t + \epsilon_{it}\end{aligned}$$

There is no well-defined control. Rather, the *Low Poverty* group acts as the comparison group.

Parallel Trends

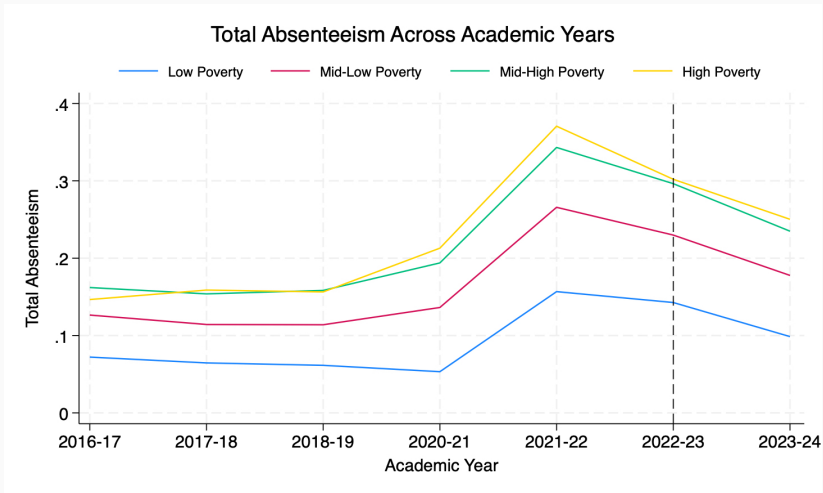


Figure 1

Results

Table 3: Total Absenteeism

	(1)	(2)	(3)	(4)
Post	0.0658*** (0.00449)	0.0660*** (0.00450)	0.0770*** (0.00471)	0.0799*** (0.00475)
Mid-Low Poverty	-0.00386 (0.00666)	0.00832 (0.00791)	-0.00937 (0.00529)	-0.00194 (0.00597)
Mid-High Poverty	-0.0311*** (0.00699)	0.0136 (0.00997)	-0.0300*** (0.00618)	-0.000523 (0.00796)
High Poverty	-0.0823*** (0.00752)	-0.00114 (0.0123)	-0.0637*** (0.00764)	-0.00595 (0.0110)
Post · Mid-Low Poverty	-0.00159 (0.00663)	0.00144 (0.00665)	-0.000248 (0.00587)	0.00203 (0.00589)
Post · Mid-High Poverty	-0.0102 (0.00546)	-0.00617 (0.00539)	-0.00796 (0.00520)	-0.00494 (0.00517)
Post · High Poverty	-0.0214*** (0.00582)	-0.0176** (0.00576)	-0.0232*** (0.00554)	-0.0203*** (0.00550)
District Fixed Effects	No	Yes	No	Yes
Year Fixed Effects	No	No	Yes	Yes
Observations	6804	6804	6804	6804
R-squared	0.1615	0.107	0.3187	0.523

Clustered by district. Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Chronic Absenteeism by Gender

	(1) Female	(2) Male
Post	0.0798*** (0.00484)	0.0818*** (0.00462)
Mid-Low Poverty	-0.00236 (0.00601)	0.00401 (0.00604)
Mid-High Poverty	-0.000117 (0.00824)	0.00398 (0.00806)
High Poverty	-0.00405 (0.0121)	-0.00275 (0.0105)
Post · Mid-Low Poverty	0.00130 (0.00605)	0.00101 (0.00593)
Post · Mid-High Poverty	-0.00658 (0.00509)	-0.00612 (0.00514)
Post · High Poverty	-0.0203*** (0.00565)	-0.0178** (0.00549)
District Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	6728	6754
R-squared	0.511	0.518

Clustered by district. Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Chronic Absenteeism by Race

	(1) Asian	(2) Black	(3) Hispanic	(4) White
Post	0.0612*** (0.00630)	0.116*** (0.00835)	0.0922*** (0.00475)	0.0733*** (0.00531)
Mid-Low Poverty	0.0164* (0.00643)	0.0198* (0.00886)	0.00204 (0.00617)	0.00101 (0.00593)
Mid-High Poverty	0.0279** (0.00949)	0.0273* (0.0133)	0.00460 (0.00840)	0.00295 (0.00888)
High Poverty	0.0273* (0.0113)	0.0253 (0.0156)	0.00147 (0.0111)	-0.00825 (0.0113)
Post · Mid-Low Poverty	-0.00668 (0.00773)	-0.0247* (0.0108)	0.000715 (0.00611)	-0.00710 (0.00656)
Post · Mid-High Poverty	-0.00966 (0.00696)	-0.0260** (0.00897)	-0.00391 (0.00548)	-0.0155** (0.00547)
Post · High Poverty	-0.0232*** (0.00660)	-0.0202 (0.0109)	-0.00952 (0.00597)	-0.0231*** (0.00573)
District Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	4365	4129	6550	6578
R-squared	0.332	0.407	0.542	0.444

Clustered by district. Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Chronic Absenteeism by Program Subgroup

	(1) Disabled	(2) Foster	(3) Homeless
Post	0.0968*** (0.00551)	0.0794*** (0.00950)	0.0945*** (0.00875)
Mid-Low Poverty	0.00538 (0.00728)	0.00388 (0.00953)	0.000578 (0.00897)
Mid-High Poverty	0.00718 (0.00961)	0.0134 (0.0153)	0.0411** (0.0143)
High Poverty	0.00316 (0.0118)	0.0000698 (0.0265)	0.0339 (0.0233)
Post · Mid-Low Poverty	-0.00488 (0.00732)	0.0182 (0.0121)	0.0126 (0.0102)
Post · Mid-High Poverty	-0.00652 (0.00612)	0.0115 (0.0113)	0.00935 (0.00991)
Post · High Poverty	-0.0119 (0.00630)	0.0637** (0.0232)	-0.00381 (0.0186)
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	6373	3676	4696
R-squared	0.528	0.281	0.336

Clustered by district. Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Summarized Results

- High Poverty districts saw the largest drop in absenteeism at about 7% from baseline ($p < 0.001$)
- Gender
 - Female: ↓ 2.03 percentage points (High, $p < 0.001$)
 - Male: ↓ 1.78 percentage points (High, $p < 0.01$)
- Race
 - Black: ↓ 2.47 (Mid-Low, $p < 0.05$), ↓ 2.60 (Mid-High, $p < 0.01$)
 - White: ↓ 1.55 (Mid-High, $p < 0.01$), ↓ 2.31 (High, $p < 0.001$)
 - Asian: ↓ 2.32 (High, $p < 0.001$)
 - Hispanic: no significant results
- Program Subgroups
 - Disabled and Homeless: reductions but not significant
 - Foster: ↑ 6.37 (High, $p < 0.01$)

Conclusion

Major Findings

- Universal Meals Program (UMP) significantly reduced chronic absenteeism, especially in high-poverty districts
- Largest effects observed among female, male, Asian, White, and Black students in high-need areas
- Foster youth did not benefit—highlighting need for targeted support beyond food access

Conclusion

- UMP demonstrates potential for improving attendance and reducing educational disparities
- Potential policy recommendation to expand universal meal programs and provide additional support services for vulnerable students
- Future research
 - Examine long-term effects of UMP on academic outcomes
 - Explore impact of UMP on other student outcomes (e.g., mental health, nutrition, graduation rate)

References

- American Federation of Teachers. (2016). <https://www.aft.org/childrens-health-safety-and-well-being/childrens-access-care/chronic-absenteeism>
- Balfanz, R., & Byrnes, V. (2012). *Chronic absenteeism: Summarizing what we know from nationally available data*. Johns Hopkins University Center for Social Organization of Schools.
- Cohen, J. F. W., Hecht, A. A., McLoughlin, G. M., Turner, L., & Schwartz, M. B. (2021). *Universal school meals and associations with student participation, attendance, academic performance, diet quality, food security, and body mass index: A systematic review*. *Nutrients*, 13(3), 911. <https://doi.org/10.3390/nu13030911>
- Deming, D. J. (2011). *Better schools, less crime?*. *The Quarterly Journal of Economics*, 126(4), 2063–2115. <https://doi.org/10.1093/qje/qjr036>
- Frisvold, D. E. (2015). *Nutrition and cognitive achievement: An evaluation of the School Breakfast Program*. *Journal of Public Economics*, 124, 91–104. <https://doi.org/10.1016/j.jpubeco.2014.12.003>
- Gordanier, J., Ozturk, O., Williams, B., & Zhan, C. (2020). *Free lunch for all! The effect of the community eligibility provision on academic outcomes*. *Economics of Education Review*, 77, 101999. <https://doi.org/10.1016/j.econedurev.2020.101999>
- Gottfried, M. A. (2014). *Chronic absenteeism and its effects on students' academic and socioemotional outcomes*. *Journal of Education for Students Placed at Risk (JESPAR)*, 19(2), 53–75. <https://doi.org/10.1080/10824669.2014.962696>
- Lochner, L., & Moretti, E. (2004). *The effect of education on crime: Evidence from prison inmates, arrests, and self-reports*. *American Economic Review*, 94(1), 155–189. <https://doi.org/10.1257/000282804322970751>
- National Center for Education Statistics. (2024). *Concentration of Public School Students Eligible for Free or Reduced-Price Lunch*. Condition of Education. U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/programs/coe/indicator/clb>
- Schwartz, A. E., & Rothbart, M. W. (2019). *Let them eat lunch: The impact of universal free meals on student performance*. *Journal of Policy Analysis and Management*, 39(2), 376–410. <https://doi.org/10.1002/pam.22175>

Thank you!

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